Philosophy of Introspection

Which sense of introspection an LLM experiment measures

Measuring AI Introspection

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Part 1

The question

Knowing your own mind

Compare two sentences:

"I am hungry"
does not seem to rest on evidence

"She is hungry"
rests on evidence

Philosophy of introspection asks what makes the first kind of knowledge different from the second. Philosophers disagree about the answer.

For a large language model (LLM), the disagreement decides what an experiment measures. Before a paper can say a model introspects, it has to say which sense of introspection its procedure is built to detect. Four philosophical positions give four senses, from weakest to strongest.

Two running examples

Concept injection

A vector for a word such as *bread* is added to the model's activations.

The model is then asked whether it detects an injected thought, and what the thought is about.

Temperature estimate

Comsa and Shanahan (2025) ask a Gemini model to estimate its own sampling temperature.

The model reasons from the text it has just produced.

Both end in a self-report. Each position in this deck asks a different question about how that self-report came about.

Part 2

A liberal definition

Three conditions

Kammerer and Frankish (2023) want a definition that also covers animals and machines:

"Introspection is a process by which a cognitive system represents its own current mental states, in a manner that allows the information to be used for online behavioural control."

Target	Nature	Function
the system's current states	*a representation*	*available for control*

A human version: you notice you are hungry, and that information is there to steer what you do next. An AI version: the definition names artificial systems that monitor their own resources, or whether they have the knowledge a task needs, as candidate cases.

How much the definition lets in

The definition is permissive on purpose:

- **States that happen to be mental.** The process must represent a mental state, but need not represent it *as* mental.
- **Available, not necessarily used.** The information has to be available for control. Nobody has to act on it.
- **Inference counts.** Interpreting yourself, or applying a theory of mind to yourself, qualifies.

Excluded	A scientist reading their own brain scan. The information is not available for online control, so it is not introspection.

Three sources of variation in a report

When someone says "I am hungry," three different things could make the report vary:



Kammerer and Frankish point out that data about responses alone underdetermine which of the three produced a given report.

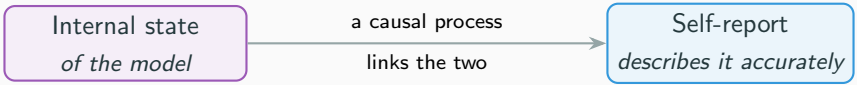
For the bread example: the report "I detect a thought about bread" could vary because the internal state varied, because the route from state to report varied, or because of how the model puts things into words.

Part 3

A causal criterion

Caused by the state it describes

Comsa and Shanahan adapt a lightweight account to LLMs. A self-report is introspective if:



- **Lightweight:** it drops the requirements that introspection be immediate or involve self-presence.
- **Internal states, not mental states:** the wording avoids implying that the model is conscious.

A human version: "I am hungry" counts if the hunger is what produced the sentence, and not, say, the habit of saying it at noon.

Creative process and temperature

They test the criterion on two cases with Gemini models:

Fails: "creative process"

The model describes how it writes. The report is better explained by mimicry of human self-reports than by a causal link to its actual states.

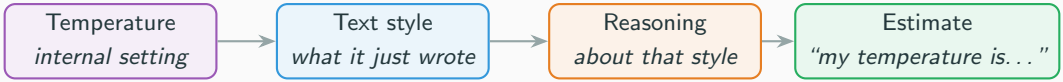
Passes: temperature

The model estimates its own sampling temperature. The temperature does shape what it wrote, and the estimate comes from that text.

The two cases are illustrations of the criterion, not a study of how accurate the model is.

The route in the temperature case

A human version: you guess how tired you were last night by rereading the messages you sent.



Every arrow is causal, so the report meets the causal criterion.

The second box is the text the model produced. Part 4 asks whether it matters that this text is public.

Part 4

Knowing it is you

Object-perception model

Shoemaker (1994) asks whether self-knowledge works like perception.

Seeing a thing

Perception gives information that identifies the object you perceive: *that* cup, over *there*.

Knowing you are hungry

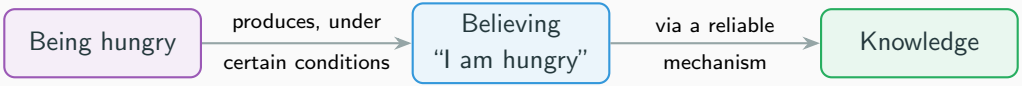
No need to identify yourself, so no room to misidentify yourself.

"if I have my usual access to my hunger, there is no room for the thought 'Someone is hungry all right, but is it me?'"

This property is called **immunity to error through misidentification** (IEM).

Reliability account

Shoemaker's alternative to the perception model:



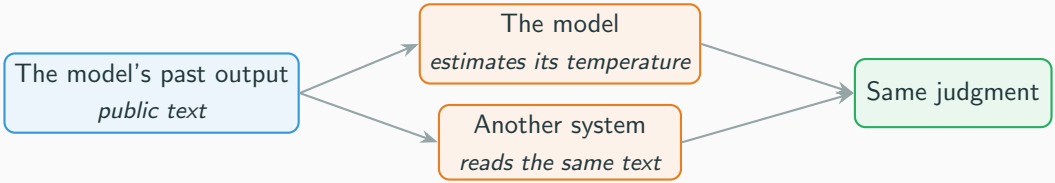
The belief counts as knowledge because the mechanism that produced it is reliable, not because of any evidence behind it.

On this account the question for an experiment is about the mechanism that links a state to the belief about it.

Self-specifying information

Zakharova (2025) grants Comsa and Shanahan their causal link, then asks whether the judgment is about the self in the right way.

IEM needs **self-specifying** information: information that leaves no room to misidentify who is in the state.



The temperature case fails this test. The model reasons from text anyone can read, so another system could reach the same judgment from it.

Self-versus-other test

A human version: does your guess about how tired you were beat a friend's guess from the same messages?

Model judges its own output vs. Model judges another model's output

Zakharova proposes this experiment: test whether models judge their own output better than other models' output.

Binder et al. run a comparison of this kind. A model introspects if it predicts its own behavior better than a second model that was trained on the first model's behavior.

Two further objections

Zakharova raises two more:

No persistent subject

Introspection presupposes a persistent subject. She argues LLMs lack one.

Too wide a net

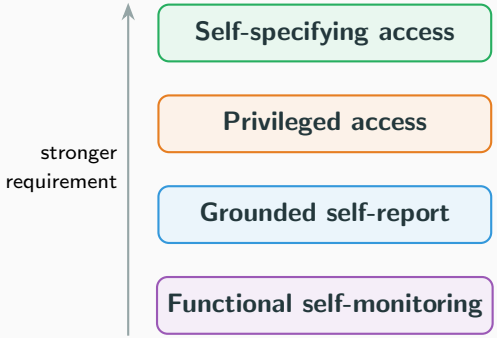
The lightweight account does not separate introspection from functional self-monitoring, which every complex system has.

The second objection gives a name, *functional self-monitoring*, to the weakest of the four senses in the next part.

Part 5

Four senses of introspection

Four senses, weakest to strongest



The report rests on information that could only be about the self. *Zakharova; Shoemaker*

The system reports on itself better than an outside observer with the same public information. *Zakharova; Binder et al.*

The report is caused by the internal state it describes. *Comsa and Shanahan*

An internal state is represented and available for behavioral control. *Kammerer and Frankish*

Concept injection is built to test the second sense. The temperature case meets the second sense and, on Zakharova's reading, fails the fourth.

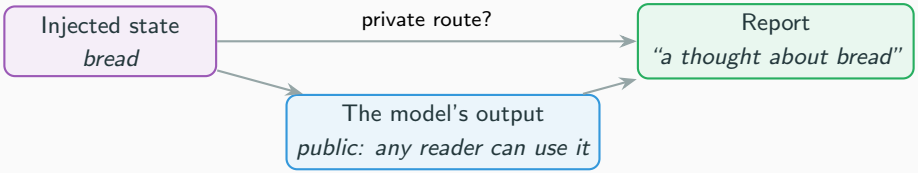
One design per sense

Each sense calls for a different experiment:

Sense	Design
Grounded self-report	Intervene on the internal state and check that the report follows. Concept injection is this design: inject *bread*, see whether the report says bread.
Privileged access	Compare the model's report about itself with another system's report from the same evidence, as Binder et al. do.
Self-specifying access	Show that the information reaching the report could not have supported a report about another system.

A route through public output

Zakharova's objection names a confound for the injection design:



The internal state can cause the report by way of the model's own output, and any reader of that output could use it too.

A causal link without control over what information reaches the reporter shows grounded self-report. It does not show privileged access.

Mimicry and the measurement model

Mimicry

Comsa and Shanahan's mimicry is an alternative explanation of the indication. In measurement terms it is a **systematic effect**. It is larger when the elicited report resembles human self-reports in the training data.

Mimicry is the reason the creative-process report fails the causal criterion.

Three terms to separate

Kammerer and Frankish's three sources of variation map onto what a measurement model of a self-report has to separate:

The internal state

The route from state to report

The verbalization

Part 6

Limits

Where the analogy stops

- **Conceptual, not empirical.** These positions fix what a measurand would be. They supply no instrument, standard or uncertainty. The other foundations on the site supply those.
- **The measurand may not exist.** If Zakharova is right that current models lack a persistent subject, no procedure measures introspection in her sense. The question for a paper then becomes which weaker sense it measured.
- **A reply on terms.** Kammerer and Frankish anticipate this objection. They answer that the term can be relabeled rather than abandoned.

Summary

Sense	Requirement	Source
Functional self-monitoring	A state is represented and available for behavioral control	Kammerer and Frankish; Zakharova
Grounded self-report	The report is caused by the state it describes	Comsa and Shanahan
Privileged access	Better on itself than an outside observer with the same public information	Zakharova; Binder et al.
Self-specifying access	The report rests on information that could not be about another system	Zakharova; Shoemaker

The measurand stage of the measurement card asks which of the four a paper tests. Mimicry and public information belong to the instrument stage, the self-versus-other test to comparability, and the existence of a subject to validity conditions.

Reading

- Kammerer and Frankish, What forms could introspective systems take? A research programme (2023): liberal working definition; three sources of variation.
- Comsa and Shanahan, Does it make sense to speak of introspection in large language models? (2025): causal criterion; creative-process and temperature cases.
- Shoemaker, Self-knowledge and 'inner sense'. Lecture I: The object perception model (1994): identification information; IEM; reliability account.
- Zakharova, Missing the subject: Introspection in large language models (2025): self-specifying information; self-versus-other test.
- Binder et al., Looking inward: Language models can learn about themselves by introspection: self-versus-other prediction design.